

ST BRANDON ILE RAPHAEL, Mauritius

WMO: 619860

Lat: 16.433S

Lon: 59.605E

Elev: 13

StdP: 14.69

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	71.2	71.9	61.4	81.3	73.9	62.8	85.5	74.3	28.4	74.0	27.4	74.2	14.9	120	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	7.4	90.9	80.8	89.8	80.5	88.8	80.0	82.1	88.6	81.5	87.8	80.9	86.9	8.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.1	156.5	86.2	79.7	154.3	85.7	79.1	151.3	85.1	45.9	89.1	45.2	88.2	44.6	87.4	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.8	26.5	24.0	69.2	93.1	1.1	1.0	68.4	93.8	67.7	94.4	67.1	95.0	66.3	95.8		
			63.2	83.6	1.9	0.6	61.9	84.1	60.8	84.4	59.7	84.8	58.4	85.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.2	83.7	83.8	84.0	82.8	80.6	77.7	76.4	75.9	76.5	78.1	80.4	82.6	
	DBStd	3.57	2.01	2.10	1.77	1.72	1.93	1.87	1.70	1.74	1.68	1.99	2.22	1.95	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	11023	1046	946	1054	985	949	832	819	802	795	873	911	1011	
	CDD65	5548	581	526	589	535	484	382	354	337	345	408	461	546	
	CDH74	49460	6753	6212	6914	5829	4534	2408	1572	1252	1476	2601	4033	5876	
	CDH80	12894	2407	2264	2518	1686	802	162	62	48	82	297	822	1744	
Wind	WSAvg	13.4	10.7	11.2	12.0	13.0	13.3	15.7	15.6	15.0	14.7	14.1	13.4	12.2	
Precipitation	PrecAvg	37.6	6.3	7.4	5.3	4.0	2.8	2.0	1.9	1.9	1.1	1.1	1.2	2.7	
	PrecMax	60.7	16.7	25.1	16.1	13.8	12.0	5.1	5.2	7.9	2.6	3.0	4.6	13.4	
	PrecMin	19.5	0.6	0.6	1.5	0.7	0.2	0.4	0.5	0.5	0.2	0.3	0.2	0.0	
	PrecStd	9.1	4.0	5.5	3.0	2.8	2.3	1.1	0.9	1.2	0.6	0.6	1.0	2.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.4	92.1	91.8	90.0	88.4	84.7	83.2	83.0	83.6	85.7	88.8	91.4	
		MCWB	80.6	81.8	81.8	80.8	79.5	76.8	75.0	74.3	74.3	76.3	78.7	80.9	
	2%	DB	90.8	90.5	90.2	88.7	86.8	83.1	81.6	81.4	82.0	84.1	87.1	89.6	
		MCWB	80.3	81.0	81.1	80.1	78.6	75.5	73.9	73.3	73.6	75.4	78.0	79.7	
	5%	DB	89.4	89.2	88.9	87.6	85.4	81.9	80.4	80.1	80.9	82.9	85.7	88.1	
		MCWB	80.1	80.6	80.5	79.6	77.8	74.8	73.3	72.4	73.0	74.7	77.3	79.0	
	10%	DB	87.9	87.8	87.7	86.4	84.1	80.8	79.2	78.9	79.7	81.7	84.4	86.8	
		MCWB	79.5	80.2	80.1	79.1	77.0	74.3	72.7	71.7	72.3	74.2	76.5	78.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.6	82.8	82.9	82.0	80.9	78.6	76.5	76.1	75.9	78.1	80.2	81.7
			MCDB	89.8	89.5	89.5	88.1	86.5	83.3	81.2	81.3	80.9	83.3	86.6	89.2
2%		WB	81.5	81.9	82.1	81.1	79.7	76.9	75.3	74.5	74.7	76.8	79.0	80.8	
		MCDB	88.4	88.2	88.3	87.0	85.2	81.1	80.0	79.7	80.2	82.0	85.0	87.8	
5%		WB	80.8	81.3	81.3	80.4	78.8	75.9	74.3	73.5	73.8	75.8	78.2	80.0	
		MCDB	87.3	87.3	87.2	86.1	84.1	80.4	79.0	78.3	79.2	80.9	83.9	86.4	
10%		WB	80.1	80.7	80.7	79.7	77.9	75.2	73.5	72.6	73.0	75.0	77.4	79.1	
		MCDB	86.1	86.3	86.2	85.0	82.9	79.7	78.1	77.5	78.3	80.0	82.8	84.8	
Mean Daily Temperature Range	5% DB	MDBR	8.2	7.5	7.4	7.2	6.3	5.5	5.5	6.2	6.5	6.8	7.5	8.1	
		MCDBR	10.2	9.4	9.2	8.9	8.3	7.3	7.1	7.8	8.1	8.3	9.4	10.2	
		MCWBR	4.4	4.1	4.2	4.3	4.2	4.2	4.0	4.3	4.2	4.1	4.3	4.3	
	5% WB	MCDBR	9.3	8.5	8.4	8.3	7.3	6.5	6.5	7.1	7.6	7.8	8.6	9.2	
		MCWBR	4.3	4.1	4.2	4.5	4.4	4.0	4.0	4.4	4.3	4.2	4.3	4.4	
Clear-Sky Solar Irradiance	taub	0.409	0.426	0.415	0.403	0.389	0.374	0.378	0.377	0.383	0.394	0.394	0.394		
	taud	2.538	2.484	2.517	2.542	2.562	2.559	2.544	2.539	2.506	2.495	2.525	2.554		
	Ebn at Noon	297	289	285	278	271	270	272	282	291	295	299	301		
	Edh at Noon	35	37	35	32	29	28	30	31	34	36	36	35		
All-Sky Solar Radiation	RadAvg	2199	2024	1992	1752	1533	1406	1459	1668	1955	2160	2311	2314		
	RadStd	167	170	83	106	81	52	31	49	49	81	71	99		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page